KINDERGARTEN · INTRO BLOCK CODING

Fix the Stack

Find the wrong block and fix it

Name: ______________________ Date: ______________________

★ I can find the block that points the wrong way and fix it.

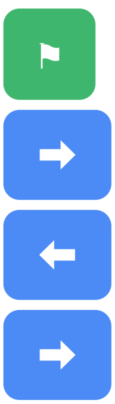
Programs can have bugs

Sometimes a program has a bug: one block points the WRONG way, so Bit misses the star ★. A coder runs the program, finds the wrong block, and fixes it. Bit starts on box 1. The star is on box 4.

Bit's stage

1 Bit	2	3	4 ★	5

Bit's program (it has a bug)



Run it from the top and see — does Bit reach the star?

1 Run it

Run the program from box 1, one block at a time. Which box does Bit land on? Is it the star? Write the box number.

2 Find the bug

One block points the WRONG way and sends Bit back. Circle the block that should not be there (the ⬅).

The fixed program

⚑

➡

➡

Draw the right arrow in the empty block.

3 You try — fix it

Draw the arrow that fixes the bug so Bit reaches the star. Then run the fixed program — which box does Bit land on now?

Big idea

Finding and fixing a wrong block is called debugging. Run it, find the bug, fix it, run it again — until Bit reaches the star.

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Quick facilitation notes & answer key

What this worksheet teaches

Children debug a stack: run it, see Bit miss the star, find the one block pointing the wrong way, and fix it. Run it, find the wrong block, fix it, run again — the debugging cycle. No coding experience needed.

Before you start (no computer needed)

No computer — paper, a pencil, and a crayon. You read each prompt aloud. Optional: paper blocks (a green flag block + arrow blocks) to stack by hand, or floor boxes the child can walk.

How to lead it (about 15 minutes)

1. Show them (you do). Run the buggy stack: "Bit missed the star — let's find the block that points the wrong way."
2. Do one together. Exercise 1 — the buggy stack lands Bit on box 2, not the star.
3. Let them try. Children circle the wrong block (Ex 2), then draw the fix and re-run (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — the buggy stack (right, left, right) from box 1 lands Bit on box 2 — not the star. Exercise 2 — the wrong block is the middle left arrow (it sends Bit back one box).

Exercise 3 — fix it to a right arrow: the stack becomes right, right, right and Bit lands on box 4, the star.

If children get stuck — and ways to adjust

Watch for: blaming the first block. Run the stack step by step and watch exactly where Bit turns the wrong way.

Make it easier: walk the buggy stack on floor boxes, then swap the wrong block and re-walk.

Make it harder: have the child make a buggy stack for a partner to find and fix.

Make it active: step out the buggy stack, then step out the fix.

Ontario Kindergarten link (official wording)

Problem Solving and Innovating — children test a sequence, identify where it goes wrong, and correct it.

In plain terms: running a stack, finding the wrong block, and fixing it — a first taste of debugging.

No reading is needed from the child — you read each prompt aloud; the child circles, numbers, or draws an arrow.

You'll know it worked when

The child finds the wrong block, fixes it to a right arrow, and lands Bit on the star.